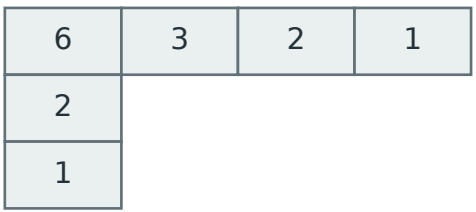
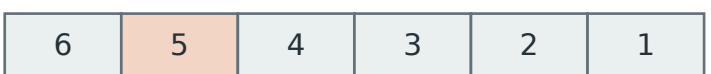


(4,1,1) is a 5-core
No hook has length 5



(6) has a hook of length 5

$$\text{At } z = 25\text{: } \prod_{h} (1 - 25/h^2)$$